

4. Sampling Distributions

While You're Waiting



A) How many ways can you partition 6 elements into 2 sets of equal size?

B) If each partition is equally likely, what is the probability of drawing one particular partition?

Causal Notation

- $Y_i(1)$:

- $Y_i(0)$:

- X_i :

- $\bar{Y}_1, \bar{Y}_0$:

- ATE:

Causal Notation

- $Y_i(1)$: the fixed value of the response of unit i after treatment
- $Y_i(0)$: the fixed value of the response of unit i after control
- τ_i : the individual treatment effect

$$\tau_i = Y_i(1) - Y_i(0)$$
- $\bar{Y}_1, \bar{Y}_0$: the average values of all $Y_i(1), Y_i(0)$
- ATE: $\bar{Y}_1 - \bar{Y}_0$
- D_i :
- Y_i :
- $\hat{\bar{Y}}_1$:
- $\hat{\bar{Y}}_0$:

$Y_i(0)$	$Y_i(1)$	τ_i
15	31	
15	22	
19	45	
2	20	
10	20	
8	15	

- $\widehat{ATE}$:

Causal Notation

- $Y_i(1)$: the fixed value of the response of unit i after treatment
- $Y_i(0)$: the fixed value of the response of unit i after control
- τ_i : the individual treatment effect
 $\tau_i = Y_i(1) - Y_i(0)$
- $\bar{Y}_1, \bar{Y}_0$: the average values of all $Y_i(1), Y_i(0)$
- ATE: $\bar{Y}_1 - \bar{Y}_0$
- D_i : a RV that is 1 if i is treated and 0 otherwise
- Y_i : a RV of the response of unit i
 $Y_i = Y_i(1)D_i + Y_i(0)(1-D_i)$
- $\hat{\bar{Y}}_1$: the average of all $Y_i(1) | D_i = 1$
- $\hat{\bar{Y}}_0$: the average of all $Y_i(0) | D_i = 0$

$Y_i(0)$	$Y_i(1)$	τ_i	D_i	Y_i
15	31			
15	22			
19	45			
2	20			
10	20			
8	15			

• $\widehat{ATE} = \hat{\bar{Y}}_1 - \hat{\bar{Y}}_0$

↳ is a RV

↳ is an unbiased estimator of ATE

$$E(\widehat{ATE}) = ATE$$

What is the Sampling Dist of $\hat{ATE}$?

Before Next Class

- Extend code to calculate all $\widehat{ATE}$ s.
- Plot their distribution
- Find $E(\widehat{ATE})$ and $V(\widehat{ATE})$.